DAY 1 · LANGUAGE MODELS · PART 1 OF 5

What Is a Language Model?

Natural language processing, tokenization, n-gram models and their limits

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Outline

- Natural Language Processing
- Text Preprocessing
- **Language Modeling**
  - **Unigrams**
  - **Bigrams**

Natural Language Processing

Natural Language Processing (NLP) is the field of study that focuses on the interaction between computers and humans through natural language, with the goal of enabling machines to understand, interpret, and respond to human language in meaningful ways.

Natural Language Processing | Applications

Text recognition:

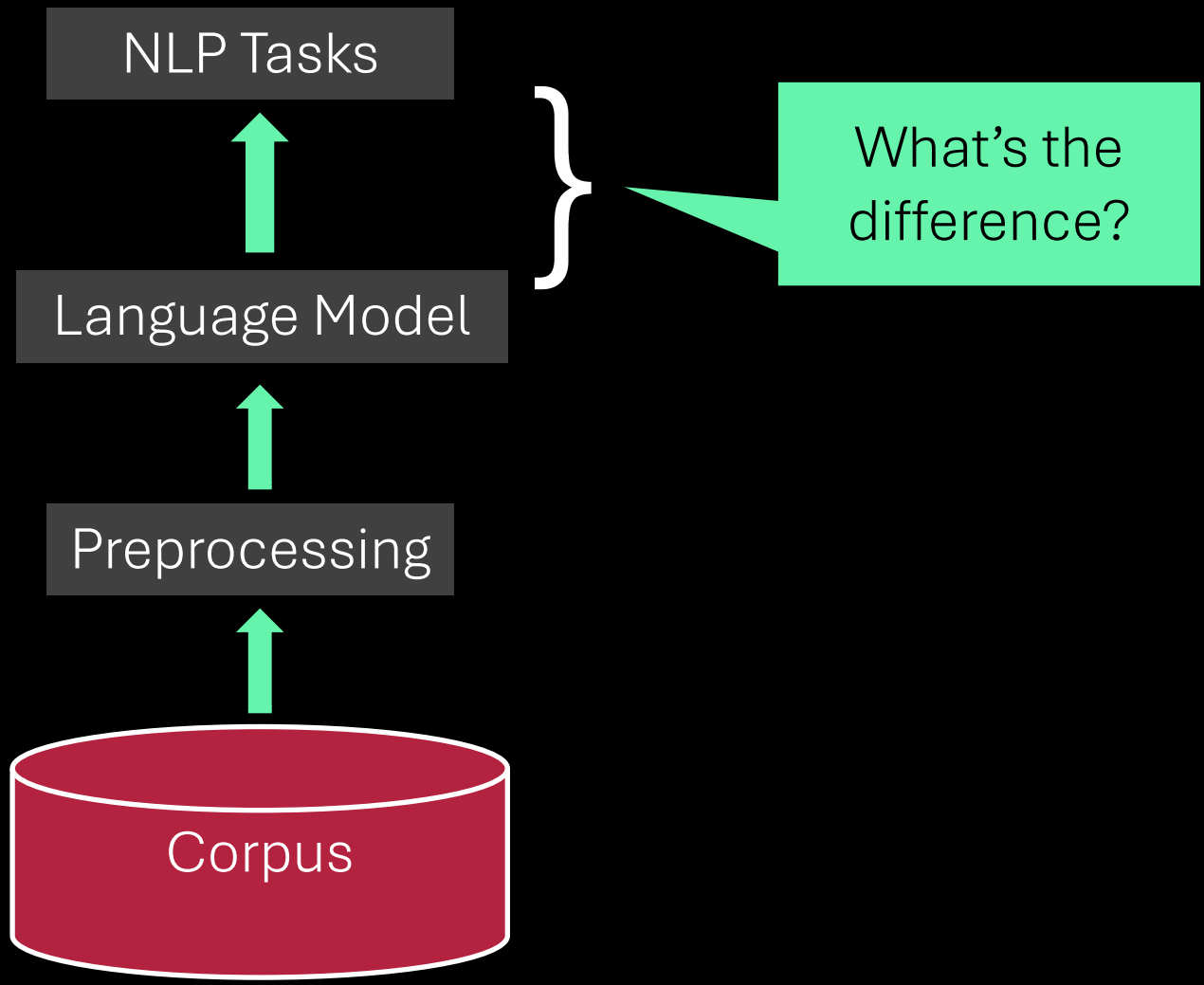


Natural Language Processing | Applications

Sentence prediction:



Pipeline


NLP Tasks
Language Model
Preprocessing
Corpus
What's the difference?

Game Time

How do we teach a toddler to answer a question? Choose the correct order.

I. **Teach the toddler to answer questions**

II. Teach the toddler language

A. I -> II

B. II -> I

C. I don’t want the toddler to answer questions

D. None of the above

NLP Task vs. Language Model

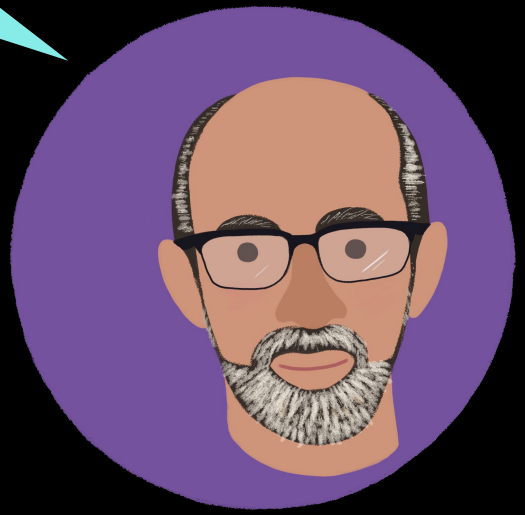
Similarly, we would want to teach our model to understand language before any downstream tasks like NER, question answering, etc.



Then what can we do about it?

We can tap into the vast ocean of unstructured text on the internet.





NLP Task vs. Language Model

3. We train the language model using the data that we got from the internet.



NLP Task vs. Language Model

3. We train the language model using the data that we got from the internet.

is the Scientific Program Director <EOS>

Language Model



Input: Pavlos is the Scientific Program Director

NLP Task vs. Language Model

4. Finally, we fine-tune our pre-trained language model to perform an NLP task, such as part-of-speech tagging.

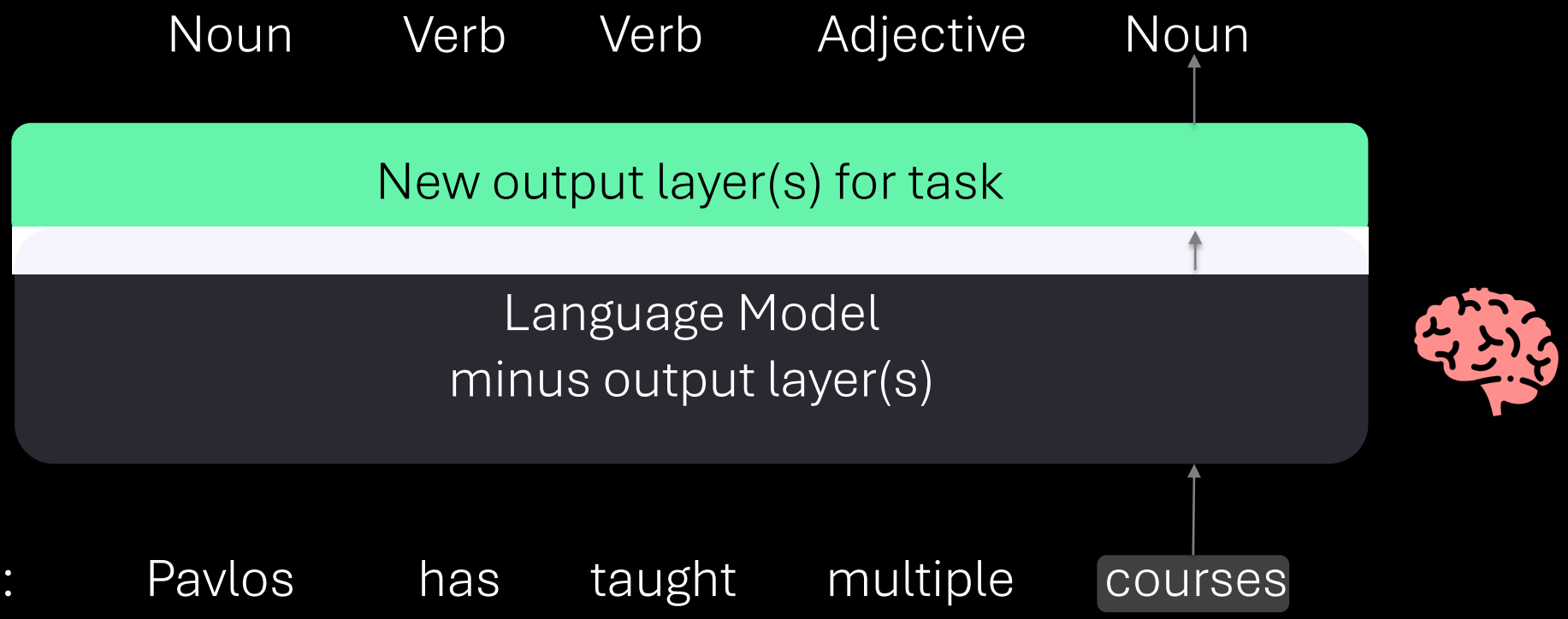
Replace the LM's output layer with a different one for our new task!



Input: Pavlos has taught multiple courses

NLP Task vs. Language Model

4. Finally, we fine-tune our pre-trained language model to perform an NLP task, such as part-of-speech tagging.



Language Modeling | Starting Point

Text is comprised of a sequence of words that convey meaning. To process it, we break the text down into smaller units known as **tokens**.

Thus, a **sentence** is transformed into a **sequence** of **tokens**.

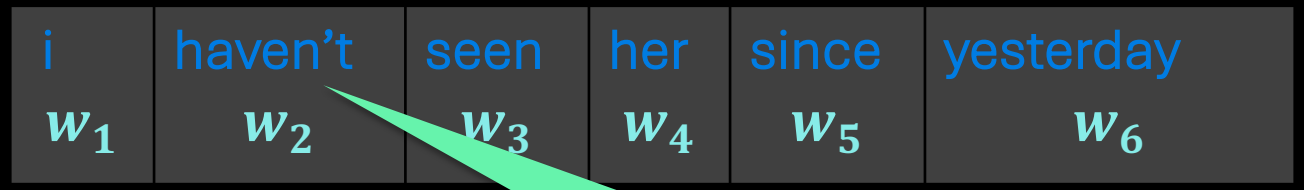
All the unique tokens from a dataset make up the **vocabulary**.

The larger the vocabulary, the larger the training set required.

We'll also see reasons to include additional 'special' tokens in our vocabulary.

Tokenization





Hyphenated phrases like "award-winning" are not split.

Usually, punctuation is removed before this stage and capitalization is set to lower case.

Conjunctions such as haven't are not split.

Tokenization

Second Approach: sub-word tokenization

Split the words on **statistically significant** fragments.

Tokens often lose their direct interpretabil**ity, b**ut the approach is more flexible.

Tokenization

Second Approach: sub-word tokenization

Split the words on **statistically significant** fragments.

How can we define a token?

Sub-word tokenization removes the **full word**, but keeps the smaller pieces that appear frequently across many words.

the	award	-	winning	actor	arrived	today
w_1	w_2	w_3	w_4	w_5	w_6	w_7

turn	the	handle	counter	clock	wise
w_1	w_2	w_3	w_4	w_5	w_6

Tokenization is an area of research in itself.

Language Modeling | Formal Definition

We model any sequential data as follows:

$$P(x_1, \ldots, x_T) = \prod_{t=2}^{T} p(x_t | x_{t-1}, \ldots, x_1)$$

We multiply a new term for every token — the joint probability is their product.

Joint distribution of all measurements

Conditional probability of an event, depends on all events that occurred before it

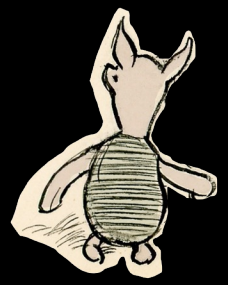
Example

The probability of the the next on-screen character can be computed as:

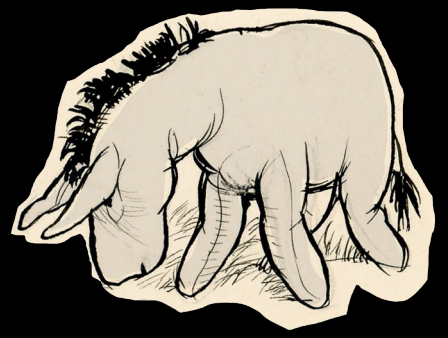
Scene 1



Scene 2



Scene 3



$$P(\text{Scene 1}, \text{Scene 2}, \text{Scene 3}) = \underbrace{P(\text{Scene 1})}_{\text{Scene 1}}\ \underbrace{P(\text{Scene 2} \mid \text{Scene 1})}_{\text{Scene 2}}\ \underbrace{P(\text{Scene 3} \mid \text{Scene 2}, \text{Scene 1})}_{\text{Scene 3}}$$

Example

Having trained a **language model** means that we can reasonably predict the behavior of the sequences.

If we have a sequence of length N, we can predict the most likely next event (i.e., sequence of length $N + 1$)

$$P(\text{  }, \text{  }, S_3) = P(\text{  }) \; P(\text{  } \mid \text{  }) \; P(S_3 \mid \text{  }, \text{  })$$

Scene 1 Scene 2 Scene 3

Language Modeling as a Sequence of Events

A **language model** estimates the probability of any sequence of words.

Let $\boldsymbol{X}$ = "**Shiv was late for class**"

w_1 w_2 w_3 w_4 w_5

$P(\boldsymbol{X}) = P(\text{"}\textbf{Shiv was late for class}\text{"})$

Language Modeling | Unigrams

How can we build a language model?

$$P(x_1, \dots, x_T) = \prod_{t=1}^{T} p(x_t)$$

Naive Approach: Unigram model

Assume each word is independent of all others.

Count how often each word occurs (in the training data).

Let $\boldsymbol{X}$ = "**Pavlos loves giving surprise quizzes**"

w_1 w_2 w_3 w_4 w_5



Language Modeling | Unigrams

How can we build a language model?

Naive Approach: Unigram model

Assume each word is independent of all others

Let $\boldsymbol{X}$ = "**Pavlos loves giving surprise quizzes**"

w_1 w_2 w_3 w_4 w_5

$\mathrm{P}(\boldsymbol{X}) = P(\textbf{Pavlos})\, P(\textbf{loves})\, P(\textbf{giving})\, P(\textbf{surprise})\, P(\textbf{quizzes})$

= 0.00015 * 0.01 * 0.004 * 0.03 * 0.0035

= 6.3×10^{-13}

You calculate and store each of these probabilities from the training corpus.

$$P(x_1, \ldots, x_T) = \prod_{t=1}^{T} p(x_t)$$

Language Modeling | Unigram Issues

Context does not affect probabilities

$P(\text{“Pavlos loves giving surprise quizzes”}) = P(\text{“quizzes loves giving surprise Pavlos”})$

Sequence generation: What's the most likely next word?

Pavlos loves giving surprise quizzes all ___

Pavlos loves giving surprise quizzes all the

Pavlos loves giving surprise quizzes all the the

Out of vocabulary words

$$P(\text{“hungry”}) = 0$$

Language Modeling | Unigram Issues

Before Smoothing :

$$P(\text{“hungry”}) = 0$$

After Smoothing :

$$P(\text{“UNK”}) = \frac{\alpha}{|W| + \alpha\,|V|} > 0$$

Smoothing allows probability of “UNK” token to be non-zero, enabling the model to predict out-of-vocabulary words.

Language Modeling | Bigrams

How can we build a language model that uses context?

Easiest Approach: bigram model

Look at **pairs** of consecutive words

$$P(x_1, \dots, x_T) = p(x_1) \prod_{t=2}^{T} p(x_t | x_{t-1})$$

probability

Let $\boldsymbol{X}$ = "Pavlos loves giving surprise quizzes"

w_1 w_2 w_3 w_4 w_5

$\mathrm{P}(\boldsymbol{X}) = P(\text{Pavlos})\, P(\text{loves}|\text{Pavlos})\, P(\text{giving}|\text{loves})\, P(\text{surprise}|\text{giving}) \times P(\text{quizzes}|\text{surprise})$

Language Modeling | Bigrams

You calculate each of these probabilities by simply counting the occurrences:

$$P(quizzes|surprise) = \frac{C(W_1W_2)}{C(W_1w)} = \frac{C(W_1W_2)}{C(W_1)} = \frac{Total\ number\ of\ \text{“}surprise\ quizzes\text{”}}{Total\ number\ \text{“}surprise\text{”}}$$

We must store all these probabilities somewhere to be accessed at inference (prediction) time.

Let $\boldsymbol{X}$ = “Pavlos loves giving surprise quizzes”

probability

w_1 w_2 w_3 w_4 w_5

$\mathrm{P}(\boldsymbol{X})$ = P(Pavlos) P(loves|Pavlos) P(giving|loves) P(surprise|giving) P(quizzes|surprise)

Language Modeling | Bigrams

- When a word is **out-of-vocabulary**, it's assigned a probability of 0, causing the entire sentence or sequence to also have **a probability of 0**.
- **More context** (like trigrams, 4-grams) is often **desired**, but sparsity is a challenge due to **infrequent** subsequences.
- As we expand the window size for context, **storage issues** arise.
- **Raw counts** fail to convey deep **semantic** relationships, such as the similarity between 'vehicle' and 'car'.